

CAPSTONE PROJECT PRE-SUBMISSION DOCUMENT

Student Success Prediction and Early Risk Detection using Learning Analytics

1. Business Problem Understanding and Objective Formulation

Business Understanding:

LearnSphere Education Technologies is a rapidly growing online learning platform offering professional certification programs, university partnerships, corporate training, and skill-development courses. Although learner enrollments continue to increase, the organization faces declining course completion rates, inconsistent learner engagement, and increasing dropout rates. Currently, learner support is largely reactive, where interventions are initiated only after learners have fallen significantly behind. However, learner attendance, assessment performance, engagement, and platform activity can potentially provide early indicators of students who are at risk. Identifying these learners early can help the organization implement timely interventions and improve learner outcomes. The provided dataset contains approximately 50,500 learner records consisting of demographic, academic, attendance, assessment, engagement, platform usage, and administrative information.

Business Problem:

The organization lacks an effective mechanism to proactively identify learners at risk of poor academic performance or course dropout, limiting its ability to improve learner retention and course completion.

Business Objectives

- Improve learner retention and course completion rates.
- Identify at-risk learners at an early stage.
- Support instructors and student success teams with data-driven insights.
- Enable proactive interventions that improve learner engagement and learning outcomes.

Analytical Objectives

- Analyze learner demographics, attendance, engagement, assessments, and platform usage.
- Develop a predictive classification model using `student_success_flag` as the target variable.
- Identify the major factors influencing learner success.
- Generate interpretable insights and business recommendations to support timely intervention.

2. Challenges and Constraints

Challenges:

- **Data Challenges:** Approximately 500 duplicate records require removal during data preprocessing. Missing values exist in variables such as `feedback_score` and `mentor_interaction_count`. The dataset contains both numerical and categorical learner attributes collected from multiple sources, requiring preprocessing before model development.

- **Business Challenges:** Early identification of at-risk learners. Providing interpretable predictions for instructors and student success teams. Converting model insights into actionable intervention strategies.

Assumptions

- student_success_flag accurately represents learner success.
- Historical learner behavior reflects future learner patterns.
- Engagement-related variables provide meaningful indicators of learner participation.

Constraints

- The dataset does not capture external personal or socioeconomic factors affecting learner performance.

3. Background Study / Literature Review

Leading online learning platforms such as Coursera and edX use learning analytics to identify learners requiring additional support and enable timely interventions. Research also shows that classification algorithms such as Logistic Regression, Decision Trees, Random Forest, and XGBoost are commonly used for student success prediction. Key findings from the literature include:

- Early identification of at-risk learners improves retention.
- Learner engagement is a strong predictor of course completion.
- Explainable models help instructors understand why learners are identified as at risk.
- Data-driven interventions improve learner outcomes.

4. Proposed Methodology / Solution Approach

The project will follow a structured Data Science lifecycle:

Business Understanding —> Data Understanding —> Data Preparation —> Exploratory Data Analysis —> Feature Engineering —> Model Development —> Model Evaluation —> Business Recommendations

- **Data Preparation:** Data quality assessment, duplicate removal (~500 records), missing value treatment (feedback_score and mentor_interaction_count), and data preprocessing and feature transformation.
- **Exploratory Data Analysis:** Analyze relationships between learner demographics, attendance, assessments, engagement, platform usage, and student_success_flag to identify important patterns influencing learner success.
- **Model Development:** Multiple classification models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost will be developed and compared to identify a suitable model for predicting student success.
- **Model Evaluation:** Evaluate the selected model and analyze the factors influencing its predictions to support business recommendations.

The proposed solution is expected to help LearnSphere identify at-risk learners early, enabling timely interventions that improve learner engagement, retention, and course completion rates.

5. Proposed Timeline

Month	Planned Activities
July	Business understanding, literature review, and dataset exploration
August	Data cleaning, preprocessing, and exploratory data analysis, Feature engineering
September	Model development & Evaluation
October	Model refinement, interpretation of results, and business recommendations
November	Documentation, GitHub repository, final report, and presentation preparation

6. Expected Deliverables / Outputs

- Exploratory Data Analysis (EDA) Report summarizing key insights from the learner data.
- Student Success Prediction Model for identifying learners at risk.
- Business Recommendations based on analytical findings to support timely interventions.
- Final Technical Report documenting the methodology, analysis, results, and conclusions.
- GitHub Repository containing source code, documentation, and project artifacts.
- Final Presentation summarizing the project outcomes.